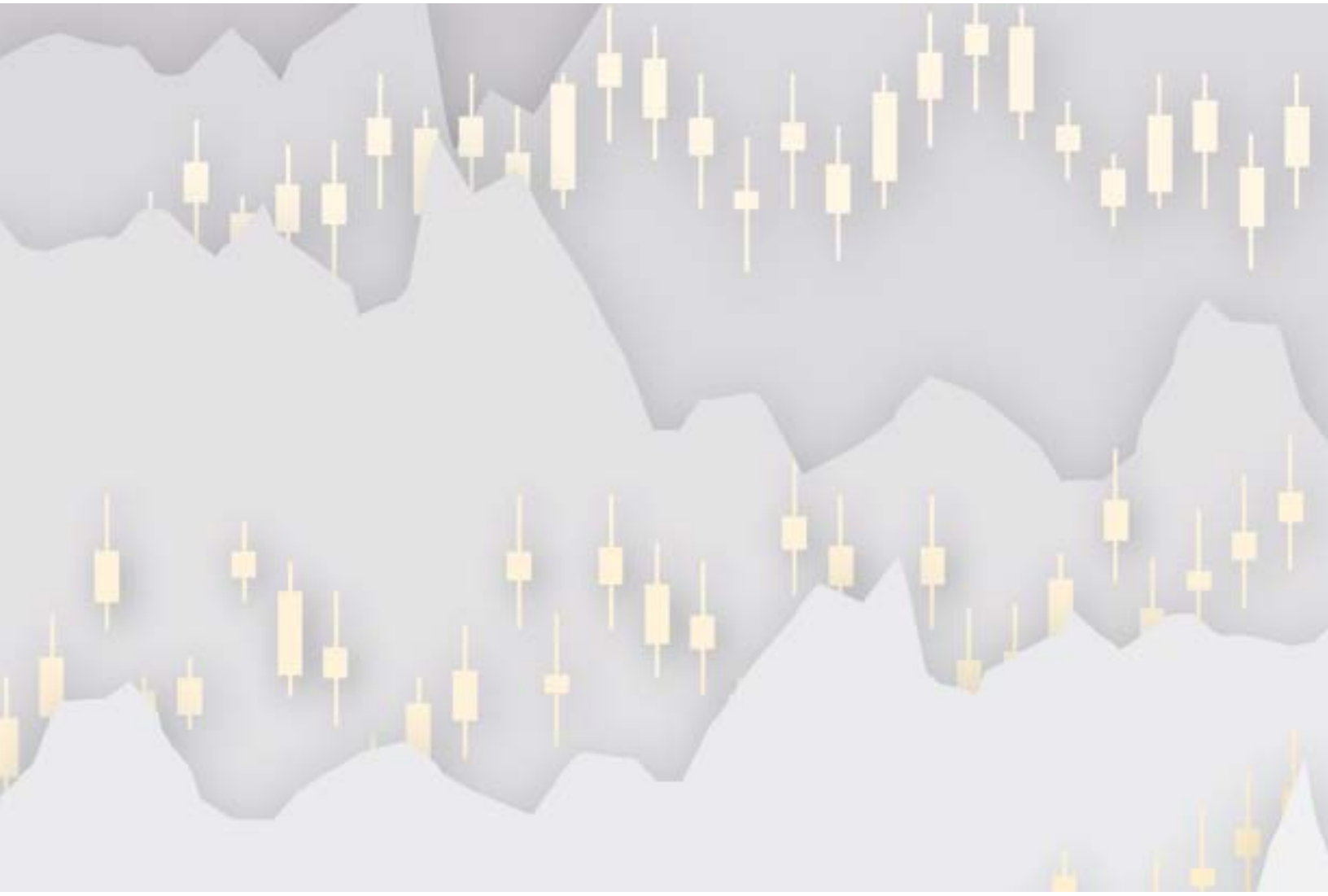
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CRYPTO LIQUIDITY 101

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INTRODUCTION

"Liquidity" is one of the more misunderstood words in crypto markets (and it has competition). Many use it to describe volume, and this makes some intuitive sense: surely, if a crypto asset has high volume, then it is liquid?

Usually, but not necessarily. In traditional markets, some assets are illiquid but have high volume, and vice versa. And in crypto assets, not all of reported volume is reliable,[1] making it even less of a useful indicator.

To compound the confusion, crypto markets have many idiosyncrasies that make liquidity an even more complicated subject than in traditional markets. The sector's retail-first evolution has led to a prevalence of small, unregulated exchanges, which dilutes volume. Also, barriers to fiat-based transactions, and the malleability of the technology, have encouraged the growth of alternative transaction mechanisms. And the market's lack of regulatory clarity, as well as its volatility, have slowed down the evolution of leverage.

In this report, we will discuss these differences and more, and look at metrics that can shed light on how crypto liquidity is evolving.

To see our other papers introducing crypto investing concepts, visit coindesk.com/crypto-investment-research. Please send feedback or comments to research@coindesk.com.

1 Nikhilesh De, "Bitcoin Futures Volume Is More Significant Than You Think, Bitwise Says," coindesk.com, March 22, 2019

WHAT IS LIQUIDITY?

The markets definition of liquidity is itself fluid. It tends to be treated as a binary concept – an asset is either liquid or it's not. To some extent, this is the case: an asset is said to be "liquid" if you can buy or sell without moving the price. But, for any asset, a large enough order will move the price, no matter how high the average daily volume. Does that mean no asset is liquid? Not quite.

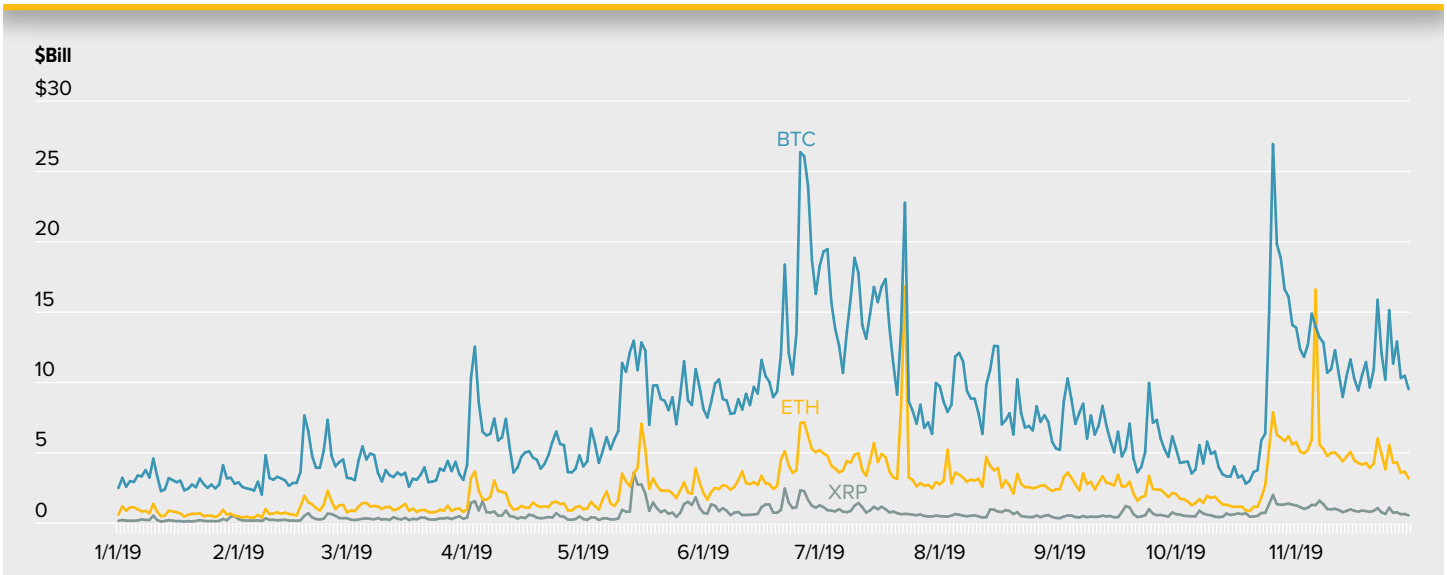
Generally, "liquid" is used to describe an asset that can be bought or sold in a reasonably large quantity without moving the price much. This makes it a relative concept, and investors tend to apply their own subjective criteria.

This is especially relevant given the relative lack of volume in most crypto assets. Bitcoin is the most liquid by far, and most crypto investors can enter and exit a reasonably sized position with relative ease. Yet, as we have often seen, a large order can whiplash the price.

The ability to enter and exit a crypto asset position without moving the price depends, as with other assets, on the size of the order – and on the chosen venue. This is where the crypto market's unusual structure makes crypto liquidity an even more complex concept than that in traditional markets.

Figure 1. Chart of relative daily volumes by asset

Crypto Assets 24-Hour Volume Vs. Time



Source: nomics.com

CRYPTO MARKET STRUCTURE

Diluted volume

The crypto market is fragmented. There are a handful of large exchanges that operate in several jurisdictions, a solid cohort of medium-sized exchanges that tend to dominate certain regions and hundreds of others that satisfy niche users – crypto data website CoinMarketCap lists over 300.

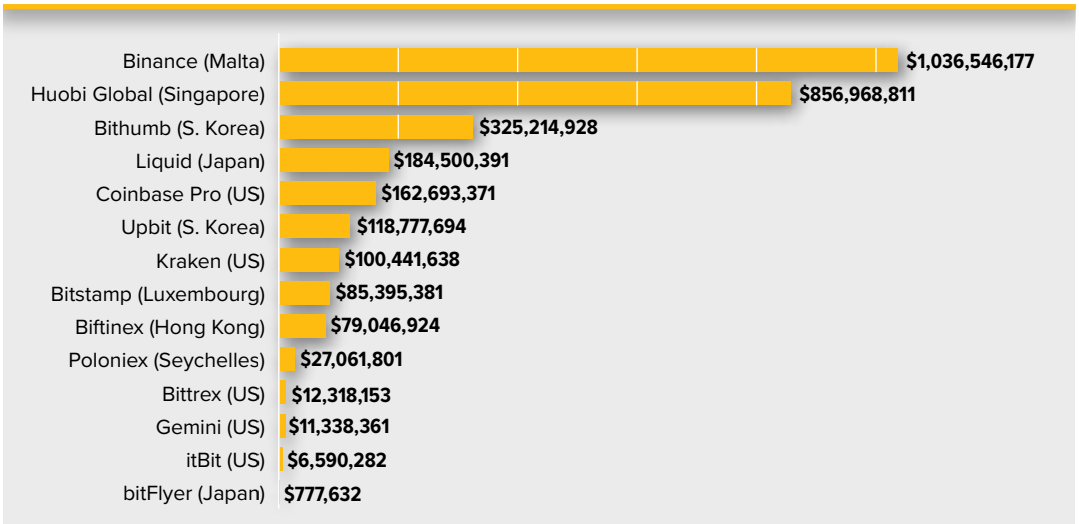
This spreads volumes across a range of venues, diluting liquidity. In spite of the wide range of choice, clients tend to choose a narrow range of venues with which to trade, since they have to fund accounts at each one. The criteria used differs for each client and for each asset – sometimes a less liquid venue is preferred because of differences in fees, rebate policies or custody solutions.

Traditional markets tend to have a narrower range of exchanges from which to choose, so there is less liquidity dilution. What's more, the choosing is not done by the end client, but by the client's broker, who will place an order at the exchange with the best price. In addition, crypto markets trade 24x7, vs the more condensed five-day-a-week opening hours of traditional exchanges. This further dilutes crypto exchange volumes, by spreading existing demand across a wider time frame.

A more diluted trading volume makes it harder to execute a large order without affecting the price. To do so, a client may need to place the order across various exchanges to avoid moving the price. This results in higher costs, which could act as a barrier to investors accumulating a large position in the first place. The order could also spook the market and encourage other investors to make trades to either protect themselves or take advantage of the movement, resulting in an unfavorable settlement price.

Furthermore, liquidity is concentrated in the top assets. It's not so much the difficulty in getting exchange listings; it's more the lack of market makers willing to invest capital in tokens that may be difficult to offload. Those that are willing, generally need to apply wide spreads to cover the risk, which reduces the smaller tokens' appeal for traders. This acts as a significant barrier to entry, which to some extent distorts the market.

Figure 2. Crypto exchanges by 24-hour volume, 30-day average
Retrieved November 25, 2019



Source: nomics.com

OTC desks

A thriving over-the-counter (OTC) market has sprung up to solve these problems, allowing investors to satisfy large orders without using exchanges. A client can place a block with an OTC desk, either voice or electronic, which will then take the other side of the trade, or find a counterparty in the market.

In this system, settlement prices tend to be more favorable, even though fees may be higher. And the buyer and seller both know the final price of the transaction before the trade is done, often useful in trading strategies and portfolio management.

The volume of OTC trading relative to exchange transactions is contested, due to the relative lack of transparency (some claim that you can calculate the breakdown by looking at blockchain wallet movements, but bundling and shared custody renders this unreliable). Nevertheless, many insiders insist that OTC volume is greater than exchange volumes:[2] earlier this year, research boutique Aite Group[3] calculated that 60–65% of all crypto trading volume occurred on OTC desks.

Lack of credit

A large part of traditional market liquidity comes from leverage, which is as yet conspicuously absent in crypto spot markets.

Often, a broker will offer credit with which clients can trade stocks, commodities, FX or bonds. This credit is often 100:1, but can be much greater.

In crypto markets, few platforms have the balance sheets required for this kind of operation. Some of the larger exchanges do offer margin trading, but the ratios are low and the resulting volumes do not yet make a significant impact.

This is largely due to the relatively small size of most exchanges, a result of the fragmented nature of the sector, and to its youth. At time of writing, very few large financial institutions from traditional markets are operating a crypto asset platform, and those that are present in the market are exhibiting sensible caution, especially when it comes to credit.

This acts as a significant liquidity constraint, and is one of the reasons many large traders prefer the nascent crypto derivatives market, where some exchanges offer over 100x leverage.

While crypto derivatives support spot volumes by offering better hedging and price discovery mechanisms, the effect is indirect.

Credit is likely to eventually become a more prominent feature in spot crypto trading, if sector consolidation and the entrance of large finance firms boosts balance sheets, supporting more lending.

2 Frank Chaparro, "Inside B2C2: The crypto market making firm that almost closed shop in 2018, and is now growing market share across the globe," theblockcrypto.com, Nov. 20, 2019

3 Aite Group, "Institutional Crypto-Trading Platforms: Blockchain Meets Block Trade," May 2019

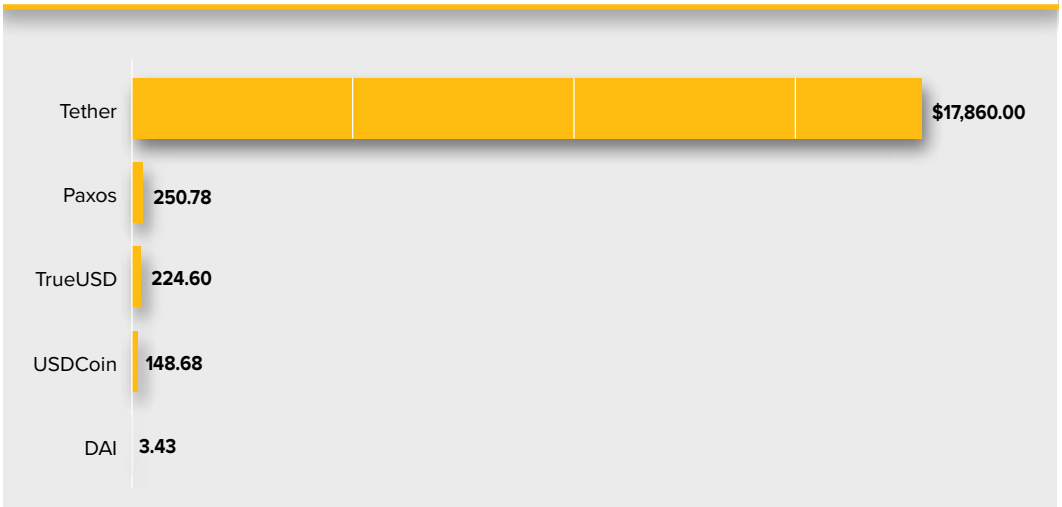
STABLECOINS

Another unusual aspect of crypto markets that affects liquidity is the proliferation of "stablecoins." These are cryptocurrencies pegged to a fixed fiat value – one unit of a dollar stablecoin, for example, would be equivalent to approximately $1.

The most liquid by far is tether, issued by a company affiliated with Bitfinex. The parent company of both is domiciled in the British Virgin Islands. Launched in 2014, tether has a significant network effect advantage which overcomes its relative lack of oversight,[4] banking problems[5] and potential regulatory issues.[6]

Figure 3. Chart market caps + volumes stablecoins

30-day average volume, November 24, 2019 (USD millions)



Source: nomics.com

Most stablecoins are based on the ethereum network, although others such as Omni Layer (which is based on bitcoin) and Stellar are also used. They move and are held in crypto wallets, just like other crypto assets, and can be bought and sold on exchanges such as Binance and Kraken.

Tether – and, to a lesser extent, its peers – enhances crypto liquidity in three main ways.

1. **Credit.** Since some platforms are willing to lend tether, it can add leverage to the sector. Users can deposit other crypto assets as collateral in exchange for a tether loan, which they can then use to buy more crypto assets, which they can use as collateral for a tether loan, and so on.
2. **Easier flow of funds between exchanges.** Fiat transfers between exchanges can be cumbersome and slow, involving fund withdrawals, bank transfers and on-ramping, all of which can take time. Since procedures vary within exchanges, banks and jurisdictions, this makes it difficult to take advantage of market movements in a timely manner, and adds a layer of friction and cost to crypto investing. Stablecoin transactions and transfers, on the other hand, are almost instantaneous, with minimal cost. Blue-chip crypto assets like bitcoin or ether are less attractive for this use, due to their price volatility.

4 Nikhilesh De, "Tether Lawyer Admits Stablecoin Now 74% Backed by Cash and Equivalents," coindesk.com, April 30, 2019

5 William Foxley, "Crypto Capital Principal Indicted on Fraud, Confirming Bitfinex Allegations," coindesk.com, Oct. 27, 2019

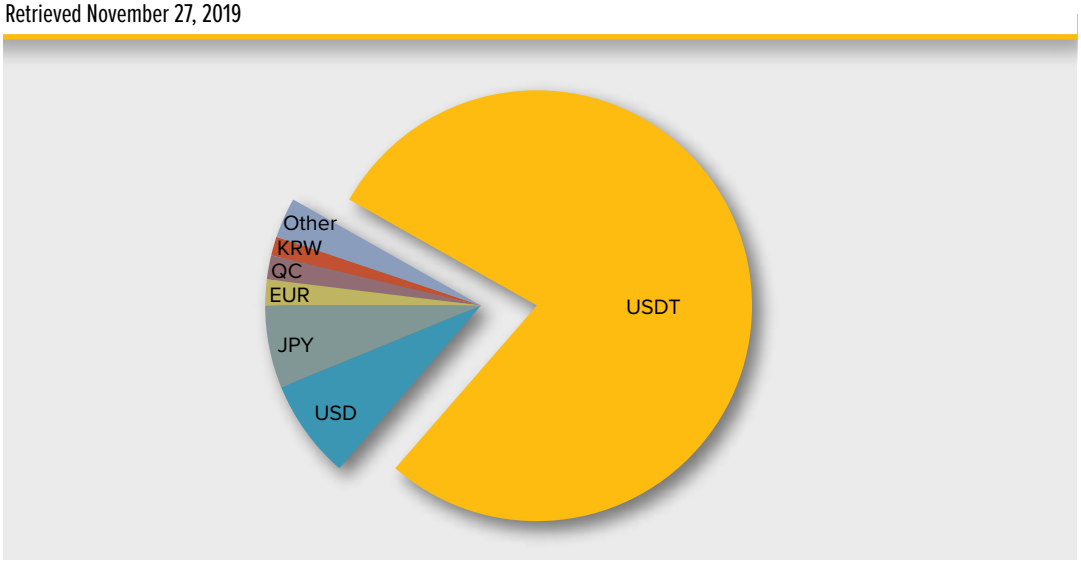
6 Nikhilesh De, "Bitfinex Doesn't Have to Compile Documents During Appeal, Judge Says," coindesk.com, Oct. 10, 2019

Several exchanges only accept stablecoins or other crypto assets; they prefer to not touch fiat, which means that their onboarding procedures are (for now) less onerous, and they don't depend on banking relationships to operate.

3. **Lower cost of capital.** The ability to rapidly move stablecoins from one exchange to another does more than facilitate the flow of funds. It also lowers the cost of capital, by removing the need to keep active balances on various exchanges. This frees up investor capital which can be put back into the sector, further enhancing volumes.

The convenience of stablecoins is such that they now account for more trading volume than any other crytocurrency,[7] and are by far the currency of choice when trading bitcoin.

Figure 4. 24-hour BTC volume by currency
Retrieved November 27, 2019



Source: cryptocompare.com

The growing role of stablecoins in crypto trading adds liquidity, but also risk. Tether in particular has come under scrutiny for its lack of rigor in verifying the token's backing. It has acknowledged[8] that its reserves are not 100%, although the latest claim at time of writing is that they currently are.[9] It is also the subject of a handful of law suits,[10] that could trigger doubt about the token's value.

A "run" on tether, with users deciding to exit the token at any cost, would most likely have a sharp short-term effect on crypto asset prices. Initially they could spike as users switch into bitcoin or ether (for example), only to subsequently fall as uncertainty regarding the market's resilience causes many investors to sell—or, general panic could cause an immediate slump across the board. Damage to confidence may have more than a short-term effect, but tether is not the only stablecoin in the market. Investors could pivot to others, which arguably have better oversight, leaving the market stronger in the long run.

7 David Johnstone, "The 'Flippening' Just Happened, Now Embrace The 'Tippening,'" medium.com, Oct. 7, 2019

8 Nikhilesh De, "Tether Lawyer Admits Stablecoin Now 74% Backed by Cash and Equivalents," coindesk.com, April 30, 2019

9 Nikhilesh De, "Tether Says Its Stablecoin Is 'Fully Backed' Again," coindesk.com, Nov. 8, 2019

10 Nathan DiCamillo, "Tether to File Motion to Dismiss Class Action Lawsuit Based on NYAG Claims," coindesk.com, Nov. 15, 2019

Or, the market could continue to not care about the actual backing of tether, preferring to rely on users' "faith" that it is worth $1. As with fiat, this continued faith would act as a strong support for deepening liquidity in the sector.

METRICS

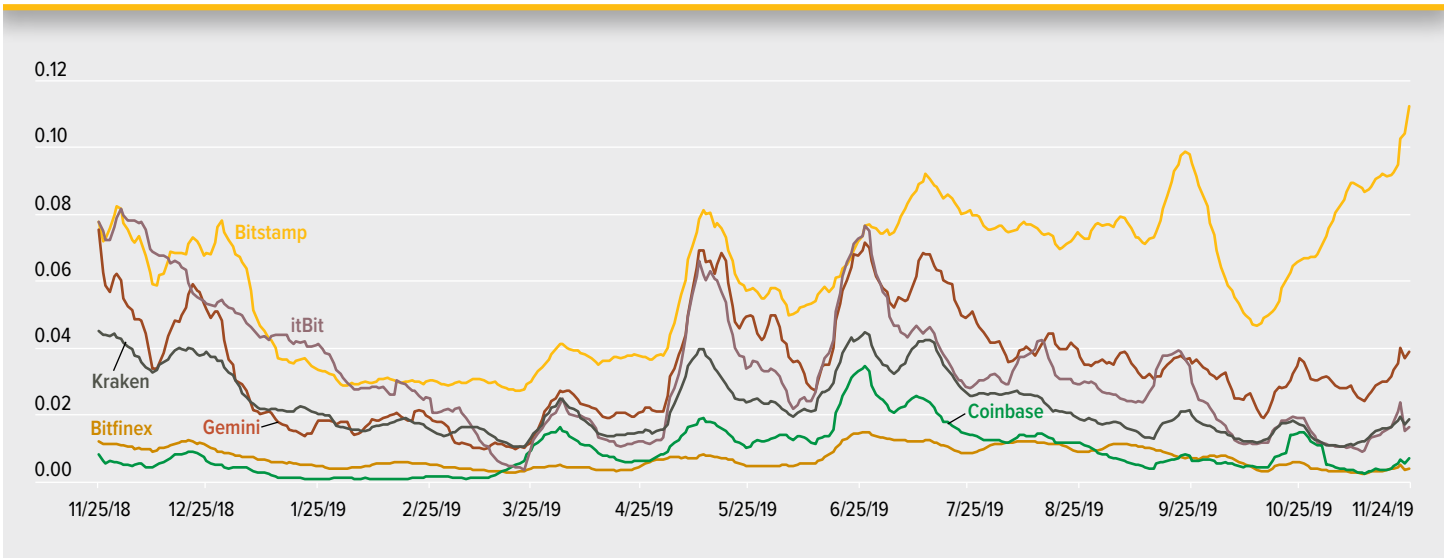
Liquidity is an elusive concept to measure, since it resides in many places simultaneously and is constantly moving.

Volume

Taking the snapshot approach, one indicator is trading volume across major exchanges. Even this has its caveats, as recent studies have shown that a significant amount of reported volume is fake, generated by wash trading and internal movements.

Several sources adjust their volume rankings and totals to account for "fake" volume. The different methodologies used lead to divergent volume figures: on Nov. 21 (taken as an example), CoinMarketCap published an unadjusted 24-hour bitcoin volume figure of $19.3bn; Nomics' reported figure was $11.5 billion, with $1.3bn "transparent volume"; Messari has 24-hour "real volume" at $273 million.

Figure 5. Cryto exchanges by 24-hour volume, 30-day average
Retrieved November 25, 2019



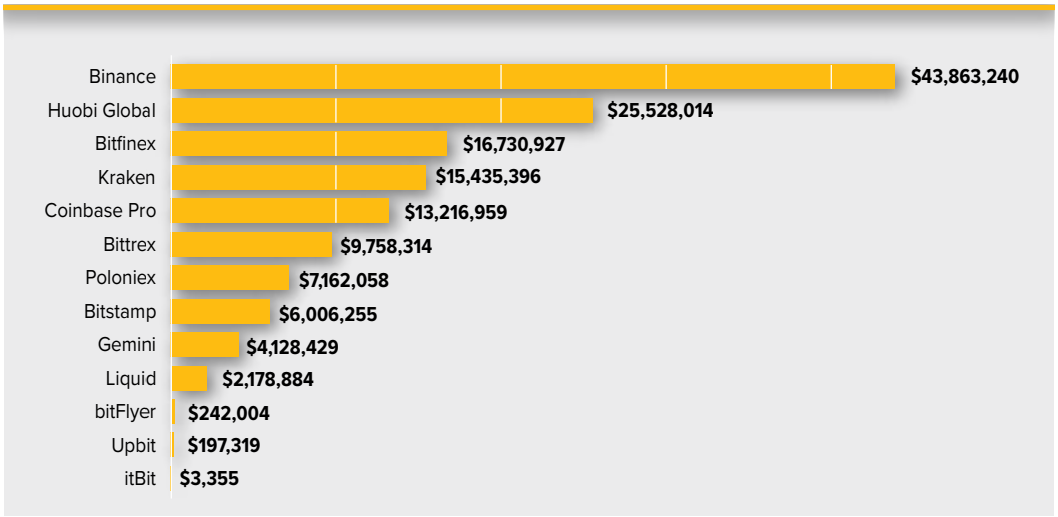
Source: nomics.com

This makes using transaction volume as a rough proxy for liquidity more complicated than in many traditional assets.

Crypto data site CoinMarketCap recently released what they call the "liquidity metric,"[11] which they hope will replace volume as the default ranking anchor for exchanges. It takes into account a wider range of variables than just volume, by looking at order books and incorporating the size of orders, their distance from the mid-price and the relative liquidity of the market pair in question.

11 CoinMarketCap, "Just Launched: New Liquidity Metric To Combat 'Volume Inflation' Problem," Nov. 11, 2019

*Figure 6. Crypto exchanges ranked by liquidity**
Retrieved November 25, 2019



*proprietary CoinMarketCap metric

Source: coinmarketcap.com

This metric has its limitations, as it could still be influenced by wash trading and other illicit practices, but it does indicate a growing awareness of the need for reliable liquidity measures.

Order book depth

Liquidity can often be indicated by an asset's order book depth. If there is solid demand and offer at stepped intervals from the mid-price, an asset can be said to be liquid. If, however, the bids and offers around the mid-price are significantly thinner than those at the outer edges, any sizeable order will significantly move the transacted price.

*Figure 7. Order Book Depth**
Retrieved November 27, 2019



Source: BTCUSD order book on Bitstamp from CryptoCompare

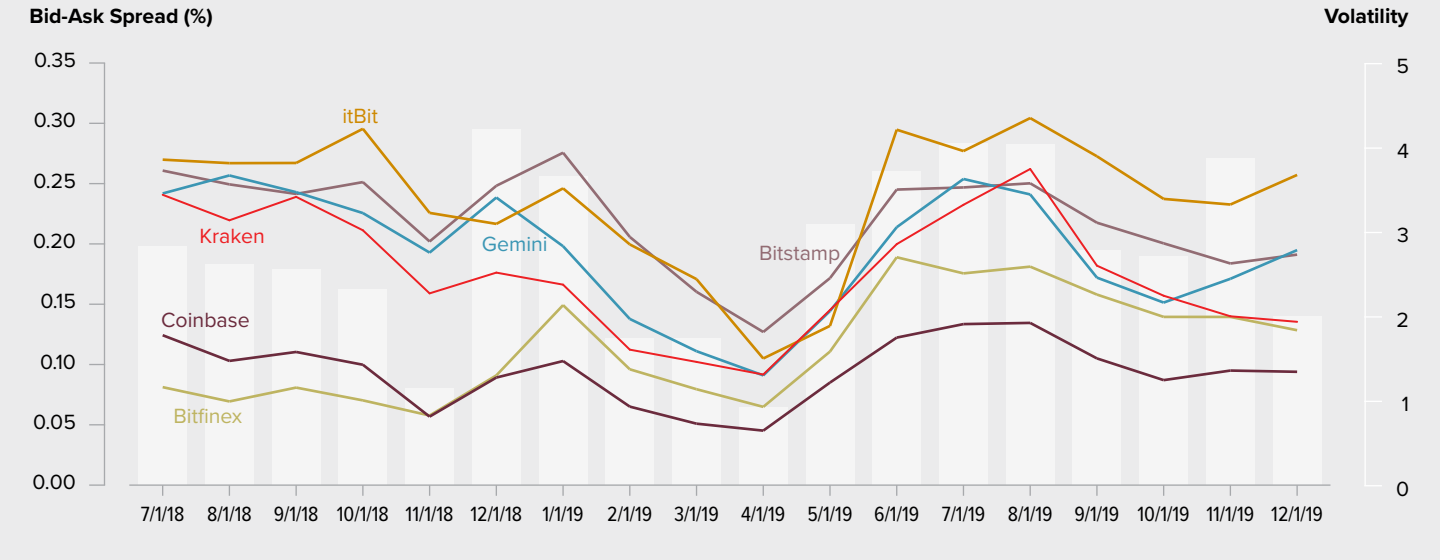
Even this is not a reliable indicator, however, given the growing presence of algorithmic trading, in which orders don't even show up unless certain conditions are met.

Spread

A more reliable indicator of liquidity in traditional markets is the average bid/ask spread. As you can see from the below chart, the spreads for the top crypto assets are low, indicating reasonable liquidity and price reliability. Wider spreads would indicate 1) a thinner market, with market makers less willing to take a position, either out of worry that volatility will move the price against them, or for fear of not being able to exit easily; and 2) a less reliable benchmark price, as an actual transaction would be materially higher or lower, which could deter investors and traders from taking a position.

Figure 8. BTC/USD bid-ask spread

10 BTC depth + Volatility, 30-day Average vs. Time



Source: bitcoinity.org

CONCLUSION

As with most aspects of crypto assets,[12] liquidity has nuances not seen in traditional markets.

The unusually fragmented structure of crypto markets thins order books, which increases the cost of trading by requiring traders to execute across a range of exchanges or by pushing large orders onto the OTC market. What's more, the lack of reliable metrics to gauge liquidity adds to investing uncertainty, especially when it comes to institutional-sized investment – yet without large market participants, attractive liquidity will be even harder to come by.

Where some see barriers, however, others see opportunity, and crypto infrastructure is evolving rapidly to overcome liquidity obstacles. Stablecoins have emerged as a potent liquidity tool, improving the flow of funds, facilitating unconventional venues and allowing a certain amount of leverage. Going forward, market consolidation and structural shifts, such as the growing use of prime brokers, as well as enhanced data reporting and metrics analysis, should continue to contribute to an overall liquidity improvement.

The sector could use it. Some metrics reveal a recent decline in market liquidity. Asset volumes are trending lower; bitcoin bid-ask spreads have been shown to be widening[13]; and bitcoin balances held by exchange wallets are now measured at rates not seen since before the 2017 peak.

It's not certain, however, how much of this trend is due to the crypto bear market, which many believe is nearing an end,[14] and how much is natural for a nascent asset class. Renewed mainstream interest in crypto assets, perhaps triggered by the upcoming "bitcoin halving"[15] or by the emergence of liquid derivative markets for a broader range of investors, could introduce fresh demand, accelerate the compression of spreads and trigger the introduction of further liquidity-enhancing measures. What's more, growing liquidity generally tends to reduce volatility, which in turn favors more liquidity.

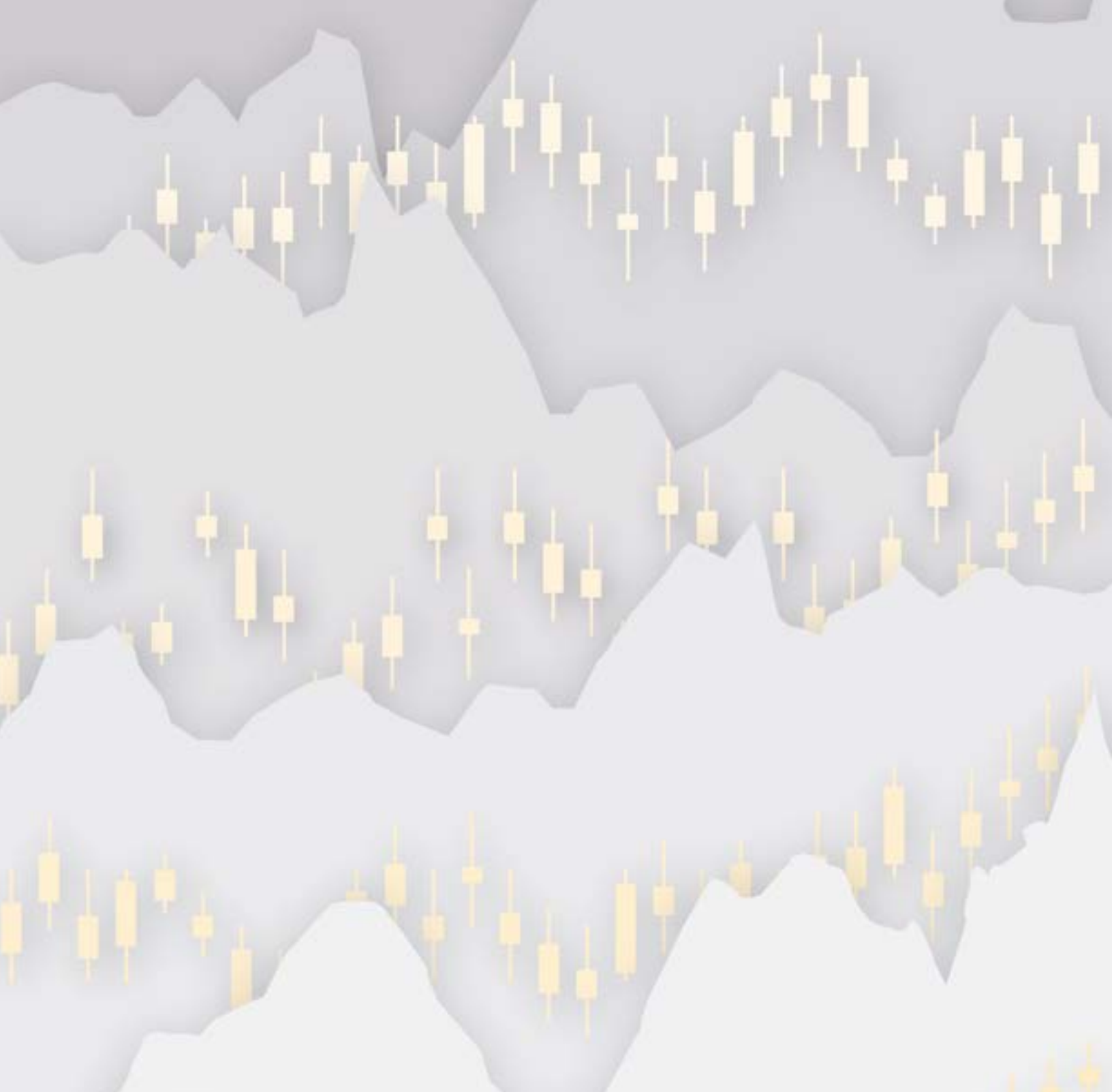
What is clear from observing both the metrics and the emerging market structure is that the sector is still in very early stages. Better information and education about the barriers and opportunities will enhance understanding not only of crypto assets, but also of markets more broadly.

12 Galen Moore and Noelle Acheson, "Crypto in Context," CoinDesk Research, July 2019

13 Galen Moore, "Bitcoin Volatility is up, LIquidity Stagnant," coindesk.com, December 7, 2019

14 Omkar Godbole, "Bear Breather? Bitcoin Looks Oversold After 50% Price Drop Since June," coindesk.com, Nov. 25, 2019

15 Noelle Acheson, "Why Bitcoin's Next 'Halving' May Not Pump the Price Like Last Time," coindesk.com, November 2019



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